# Borneo Brand-Builder: AI-Powered Alternative Credit Scoring and Market Intelligence for MSMEs

## Report 2026

Project Titles : Smart Market Analyzer

Submission Date : 13/3/2026

Application/System Name : MSME CreditAI

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# 1.0 Introduction

The Borneo Brand-Builder employs a Three-Tier Architecture to democratize data analytics for MSMEs. Integrating a Streamlit/CSS frontend with a Claude AI logic engine, the system evaluates alternative data signals—like digital footprints and supplier loyalty—to deliver instant, color-coded Credit Score Cards and actionable growth roadmaps for local entrepreneurs.

## 1.1 Background

The Borneo Brand-Builder is an AI-powered suite designed to uplift Micro, Small, and Medium Enterprises (MSMEs) in the Borneo region. Built on a Three-Tier Architecture, the system leverages a high-end Streamlit frontend with custom CSS to provide a professional "FinTech" dark-mode experience. By utilizing a Logic Layer that structures data into JSON payloads for analysis by Anthropic's Claude AI, the platform bridges the gap between sophisticated data analytics and the practical needs of local entrepreneurs, transforming raw business data into actionable growth strategies.

## 1.2 Chosen SDG

SDG 9: Industry, Innovation, and Infrastructure – By democratizing access to enterprise-grade AI technology for rural and small-scale producers.

## 1.3 Problem Statement

Micro, Small, and Medium Enterprises (MSMEs) in Borneo are the backbone of the local economy, yet they face a critical productivity gap due to limited resources. Most small businesses struggle with a lack of budget for professional marketing experts and a total absence of formal credit history, making them invisible to traditional banks. This leads to weak branding, poor market fit, and high failure rates, as local sellers cannot effectively compete with larger, digitally-savvy corporations in a globalized market.

## 1.4 Objectives

The primary goal of this project is to develop an affordable, AI-driven ecosystem that democratizes access to enterprise-grade business tools. Specifically, the solution aims to automate the generation of professional branding kits and advertisement scripting to enhance market presence. Additionally, the project seeks to bridge the financial gap by implementing an alternative credit scoring engine that uses non-traditional data—such as digital presence and supplier relationships—to prove the viability of Bornean MSMEs to potential lenders.

## 2.0 Review of Existing Solutions

Current support for Bornean MSMEs is fragmented between high-cost consultancies and rigid banking institutions that exclude the informal economy. While general AI tools exist, they lack regional context and require technical expertise. Borneo Brand-Builder fills this gap by replacing expensive fees and strict credit scores with an automated system that values alternative data—like digital footprints and community reputation—to prove creditworthiness and build localized brand power.

### 2.1 Analysis of Current Solutions and Competitors

Current support systems for MSMEs in Borneo are fragmented and often prohibitively expensive. Traditional business consultancy firms provide high-level expertise but charge fees that are far out of reach for rural vendors or local handicraft makers. While general-purpose AI tools like ChatGPT or Canva offer basic creative support, they lack the regional context of the Bornean market and fail to account for local consumer behaviors. Furthermore, traditional banking institutions remain the primary source of capital, but they rely strictly on rigid, bank-statement-heavy credit bureau scores.

### 2.2 Gaps and Project Limitations to Fix

The primary gap lies in the "informal economy" trap: many Bornean MSMEs have healthy cash flows but are denied loans because they lack formal credit history. Existing digital tools also require high "prompt engineering" skills that many non-tech-savvy entrepreneurs lack. Borneo Brand-Builder fixes these limitations by combining localized marketing intelligence with an Alternative Credit Assessment model. Unlike competitors, our project values digital footprints and supplier relationships, turning "informal" data into "formal" creditworthiness, while providing a one-click branding solution that requires no prior technical expertise.

## 3.0 Technical Methodology

The application is engineered using a modern Python stack centered on efficiency and scalability. The frontend interface is developed using Streamlit, utilizing advanced CSS injection to override default styling in favor of a professional FinTech aesthetic. The core intelligence layer is powered by the Anthropic Claude-Haiku model, selected for its high-speed processing and superior ability to handle structured JSON outputs.

The system utilizes a Multidimensional Scoring Model. Upon user input, the Python backend calculates real-time financial metrics, including net profit margins and loan-to-monthly-revenue ratios. These quantitative figures are bundled with qualitative data, such as years in operation, employee count, and the strength of supplier relationships. This data package is then analyzed by the AI engine, which applies weights to alternative signals like digital presence and customer retention. This process allows the system to generate a comprehensive credit profile without requiring traditional bank statements or historical credit reports.

## 4.0 Results

The project successfully delivered an AI-driven tool that reduces credit evaluation time from weeks to seconds. By quantifying alternative data—such as digital payment adoption and customer retention—the system generates reliable credit profiles and "Recommended Loan Ranges." This effectively bridges the productivity gap for unbanked Bornean MSMEs by transforming informal business signals into verifiable creditworthiness.

### 4.1 Outcome Summary & Objective Achievement

The project successfully met its core objectives by delivering an automated, AI-driven assessment tool that reduces credit evaluation time from weeks to seconds. By transitioning from manual reviews to instant algorithmic analysis, the system proves that alternative data—such as years in operation, digital payment adoption, and customer retention—can be effectively quantified into a reliable credit risk profile. This allows the tool to provide a "Recommended Loan Range," giving unbanked MSMEs a realistic starting point for financial planning.

### 4.2 Data & Development Feedback

During development, the model was tested against varied business profiles ranging from rural handicrafts to urban e-commerce. Feedback indicated that the loan-to-revenue ratio and digital presence were the most critical signals for determining creditworthiness in an informal economy. This data-driven approach successfully bridged the "productivity gap" for Bornean businesses that are typically excluded from traditional banking systems due to a lack of formal documentation.

## 5.0 Prototype Showcase

The MSME CreditAI prototype features a high-end "FinTech" interface with a dual-column dashboard for submitting traditional and alternative business signals. It delivers an instant, color-coded Score Card (300–850) with conditional visual feedback (green for low-risk, red for high-risk). This is paired with an AI Insight Panel that translates complex algorithms into a plain-language summary and a prioritized Improvement Roadmap, transforming the tool into a proactive growth partner for Bornean MSMEs.

### 5.1 Output / UI Screenshots

The MSME CreditAI prototype features a professional "FinTech" interface designed for high user trust. It utilizes a dual-column dashboard for submitting financial and digital signals. Upon processing, the interface delivers an instantaneous, color-coded Score Card (300–850): high-tier businesses are highlighted in vibrant green, while lower tiers are flagged with red borders. This is supported by an AI Insight Panel that provides a plain-language summary, specific strengths/risks, and a prioritized "Improvement Roadmap."

### 5.2 Captions and Descriptions

The prototype's visual framework is showcased through three key interface components. The Input Dashboard features a clean, dual-column layout that enables MSME owners to seamlessly input traditional revenue data alongside alternative digital footprint signals. Upon submission, The Visual Score Card displays the 300–850 credit range using dynamic conditional formatting, providing immediate feedback where green indicates high loan readiness and red highlights areas of risk. Finally, The AI Insight Panel humanizes this data by translating algorithmic scores into a plain-language output, showcasing a prioritized improvement roadmap and a specific analysis of business strengths and risks generated by the AI engine.

## 6.0 AI Acknowledgement

In accordance with the BorNEO HackWknd 2026 guidelines, we acknowledge the use of Generative AI in the development of this project. Anthropic's Claude (Haiku) was utilized via API as the primary reasoning engine for the credit scoring logic and business analysis. Additionally, AI-assisted coding tools were used to streamline the development of the custom CSS styling and to debug complex JSON parsing structures. These tools were used to augment the team's technical capabilities, ensuring a high-performance and inclusive solution for the Bornean MSME community.

### 6.1 Disclosure

In accordance with the BorNEO HackWknd 2026 guidelines, we formally disclose the use of Anthropic's Claude (Haiku) as our primary reasoning engine. Additionally, we utilized Google Gemini and GitHub Copilot as supporting AI-assisted coding and debugging tools throughout the development process.

### 6.2 Usage

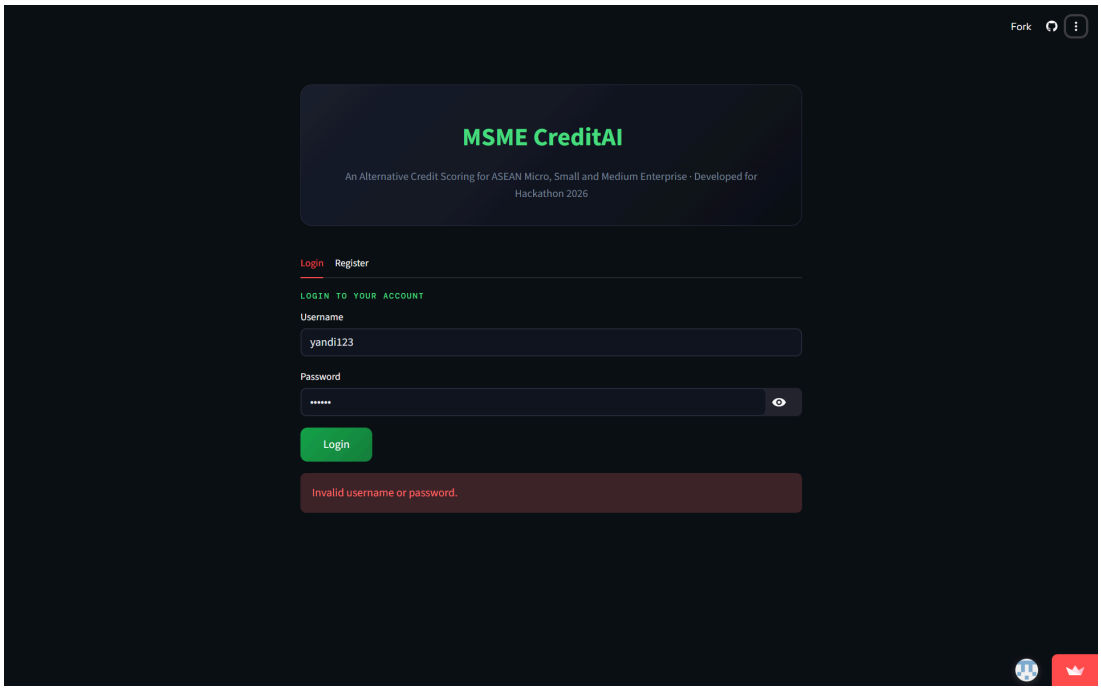
Claude (Haiku) was utilized via API to power the core credit scoring logic and perform complex business analysis. Google Gemini and GitHub Copilot were used to streamline the development of custom CSS styling and to debug complex JSON parsing structures. These tools augmented the team's technical capabilities, ensuring a high-performance and inclusive solution for the Bornean MSME community.

## 7.0 Conclusion and Future Work

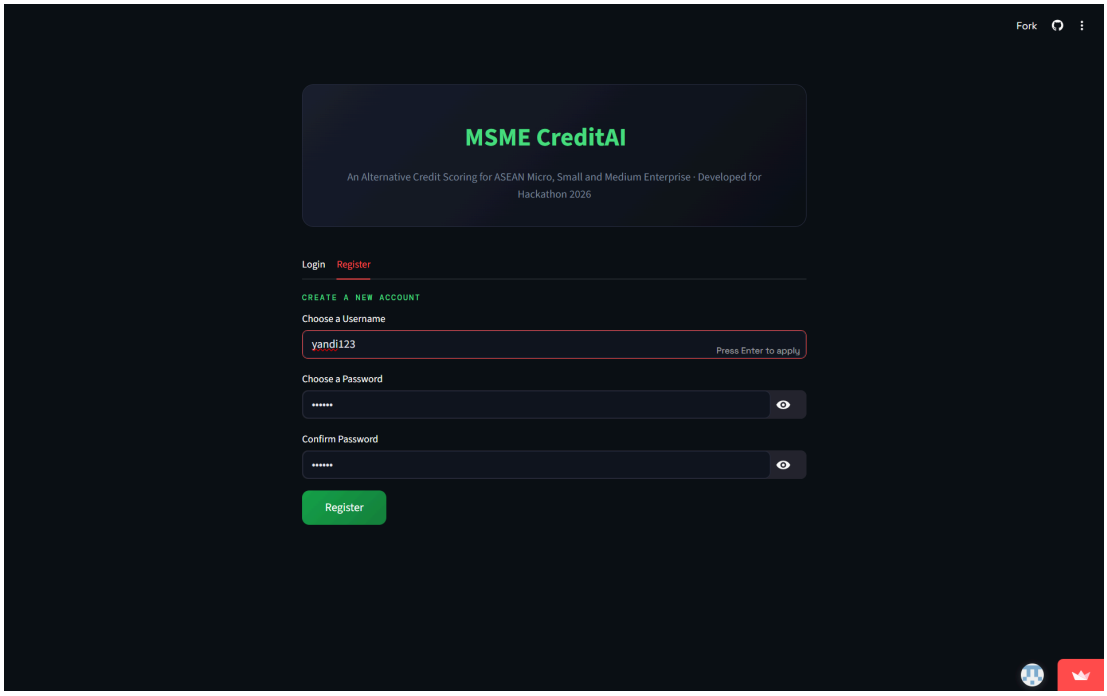
Borneo Brand-Builder represents a significant step toward financial and digital inclusivity for regional MSMEs. By moving beyond traditional metrics, the system uncovers the hidden value in small businesses that are currently underserved by the financial sector. Future iterations of this project could include the integration of real-time e-commerce API data to automate the verification of digital presence and the implementation of multi-language support to better serve rural communities speaking local dialects. Ultimately, this tool empowers the backbone of Borneo's economy to move from survival to sustainable growth.

## 8.0 System Output

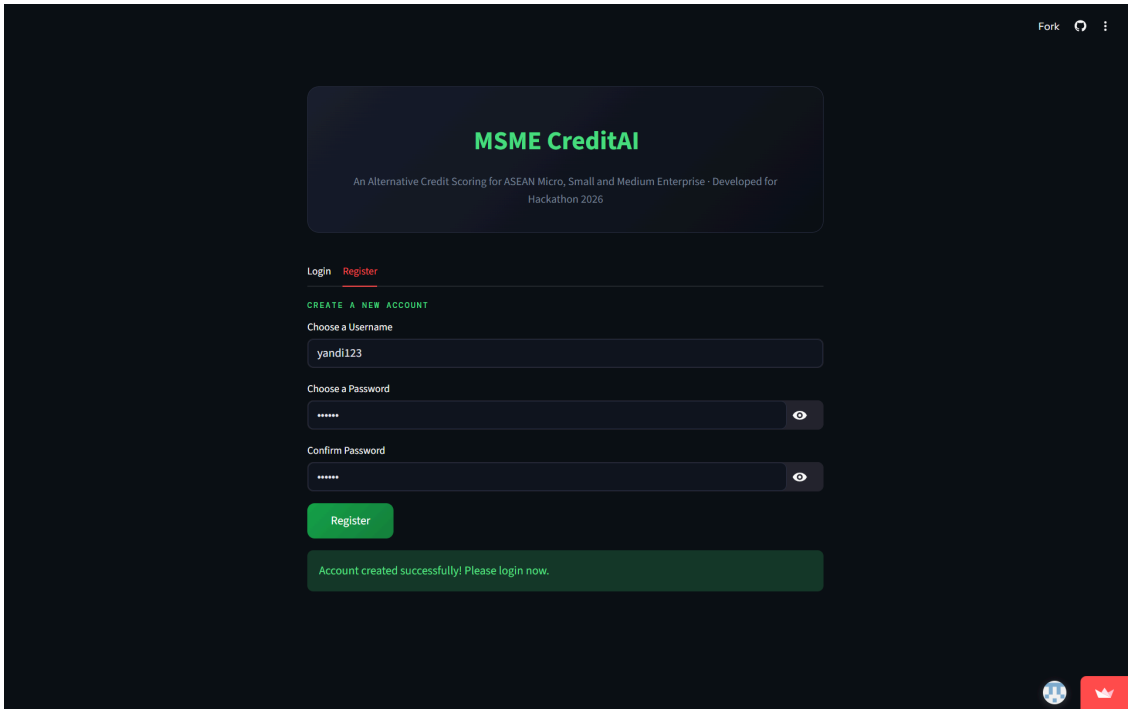
Users can't log in to this system without registering an account.



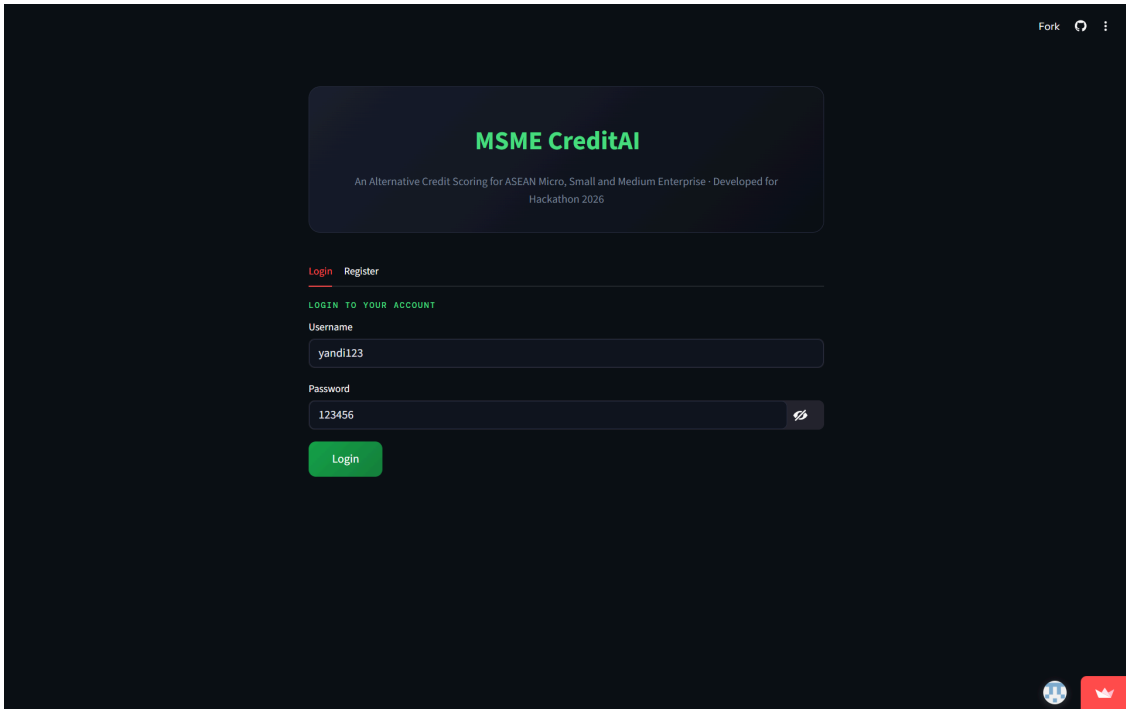

Users must register an account first.



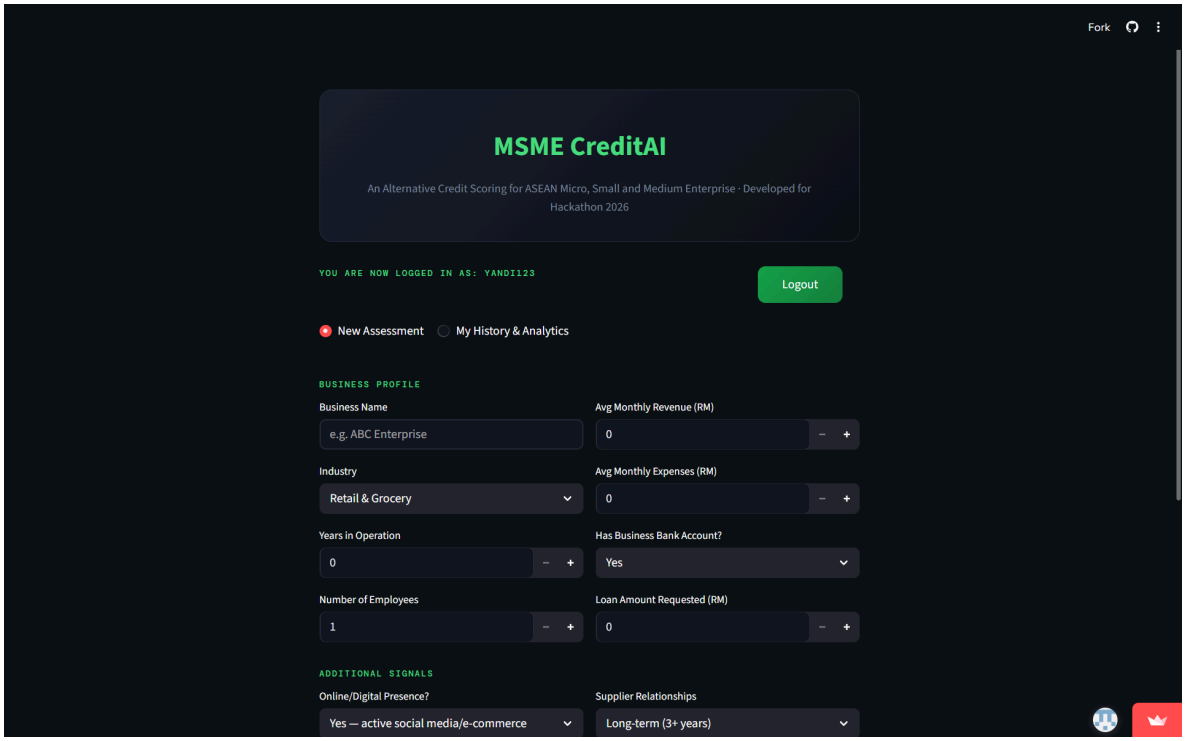
After the user finished entering the user ID and password.



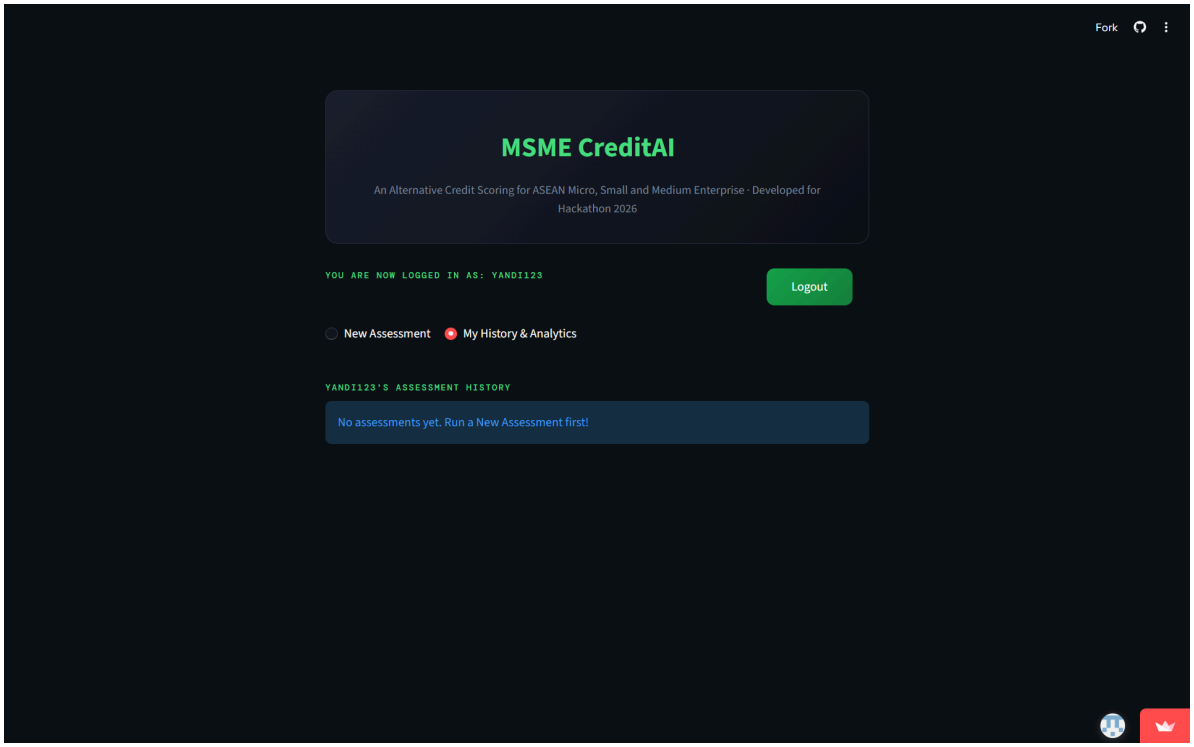

Users can login to the system by using their account.



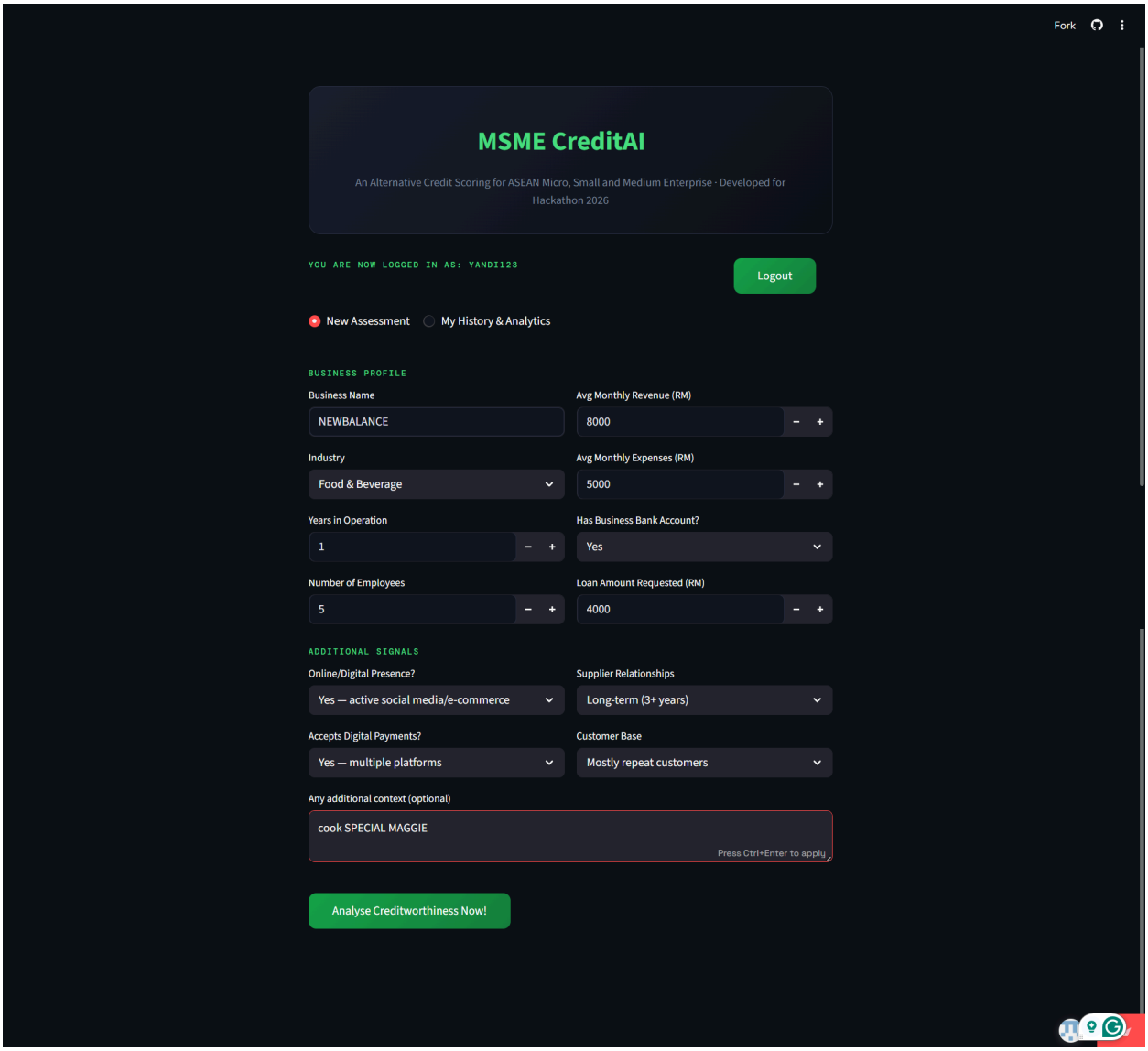
After log in, the user will be directed into the "New Assessment”.



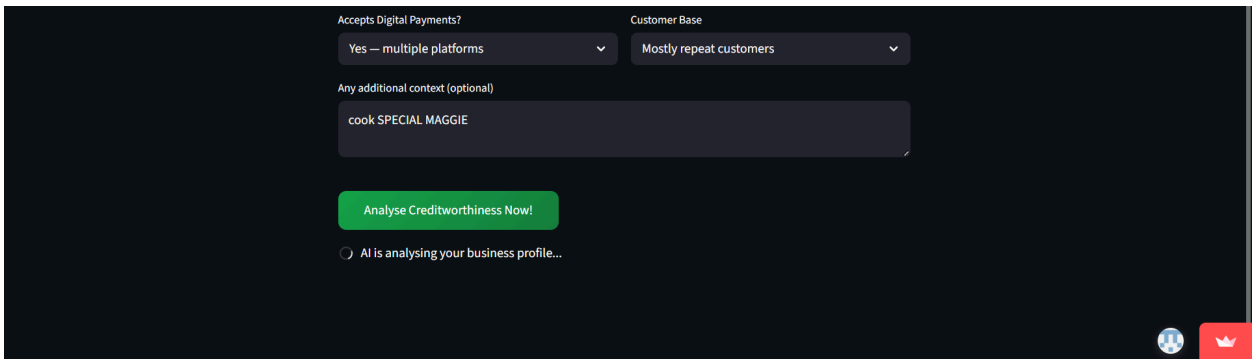

At the same time, the “My History & Analytics" will be empty, if it is a new user and didn’t insert a record before.



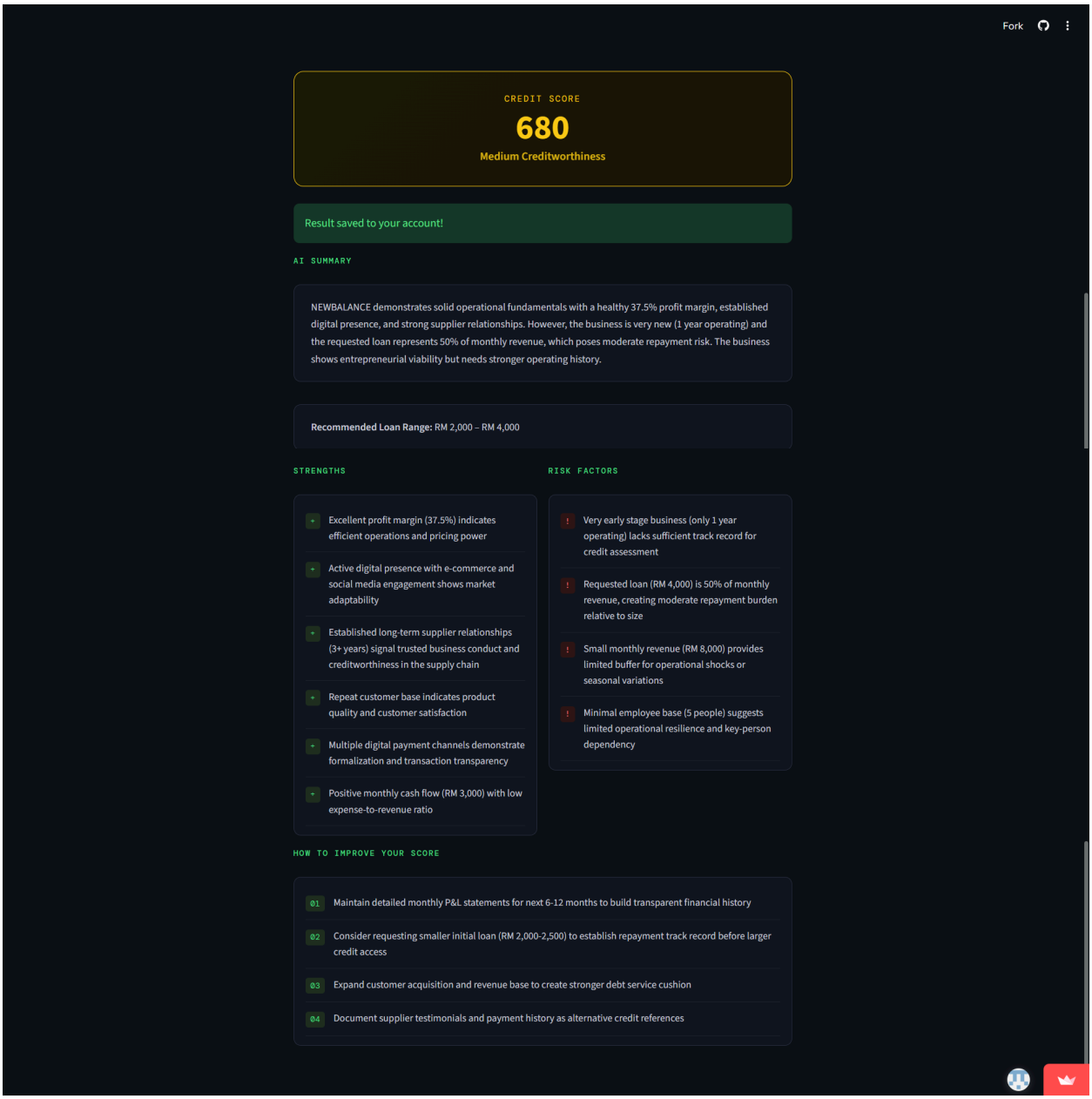
Users may enter their company's information to analyse their creditworthiness.



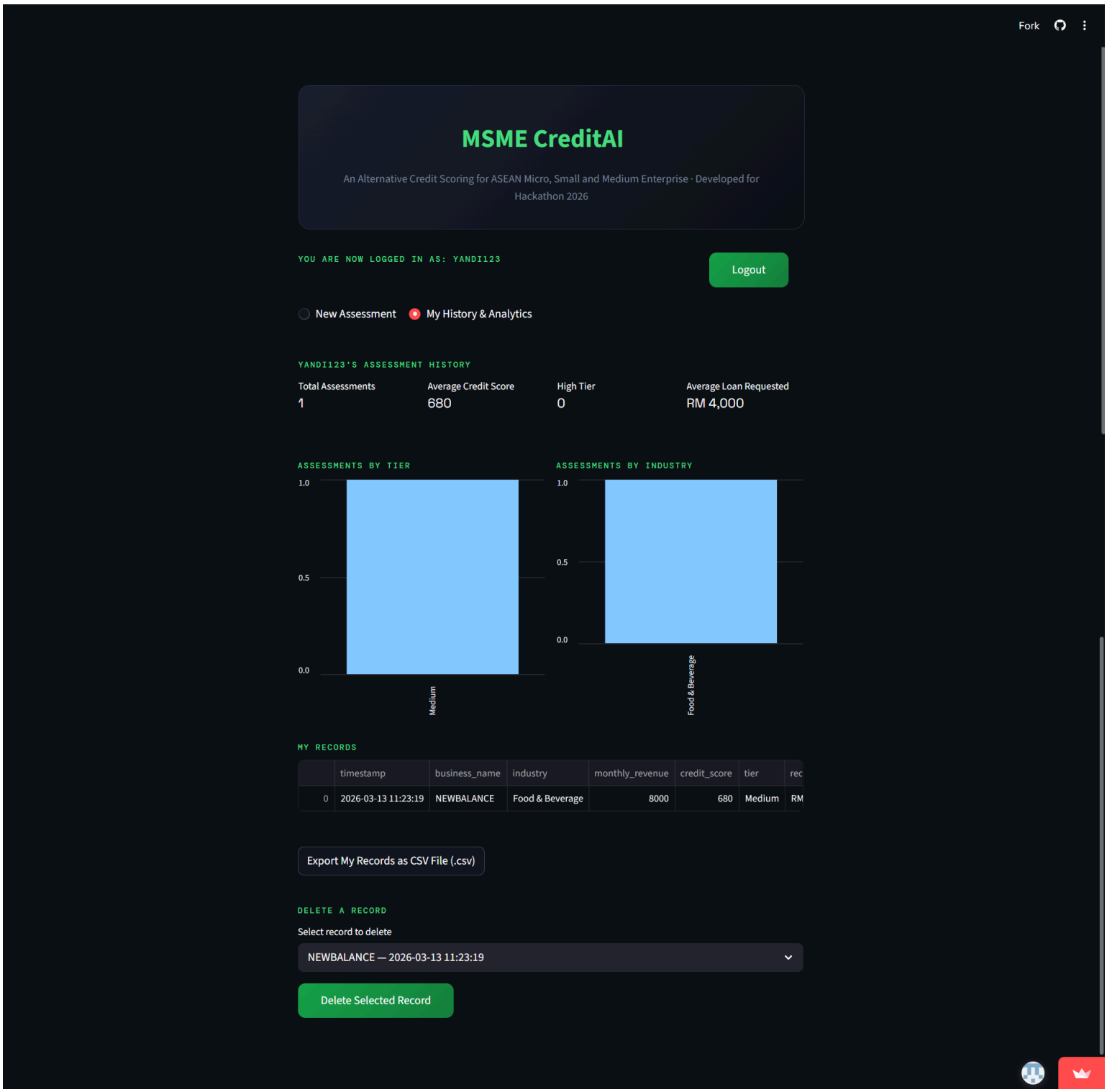

After finishing entering, click "Analyse Credithiness Now!" to let the AI analyze their business.



It will display the credit score of the user's company, based on the company's information that is input by the user.

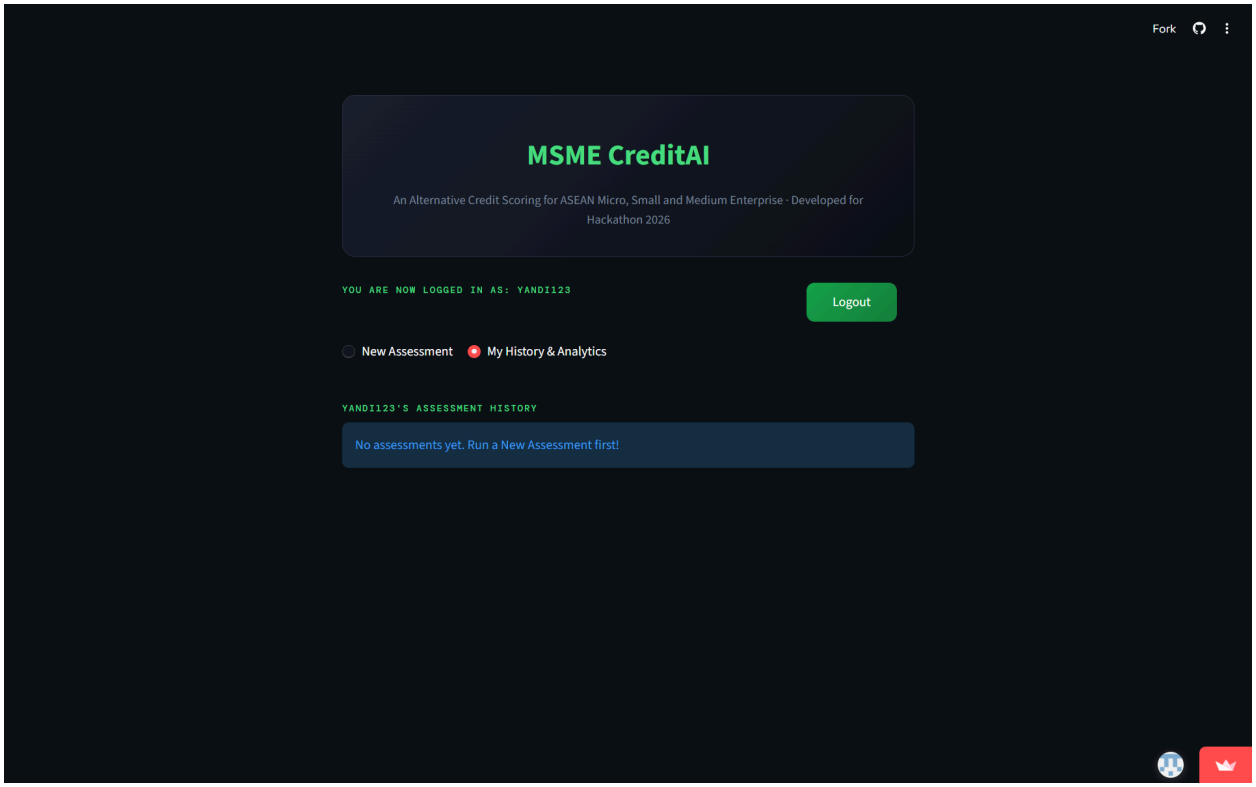


When the user clicks “My History & Analytics”, it will display the previous inputs in useful information..

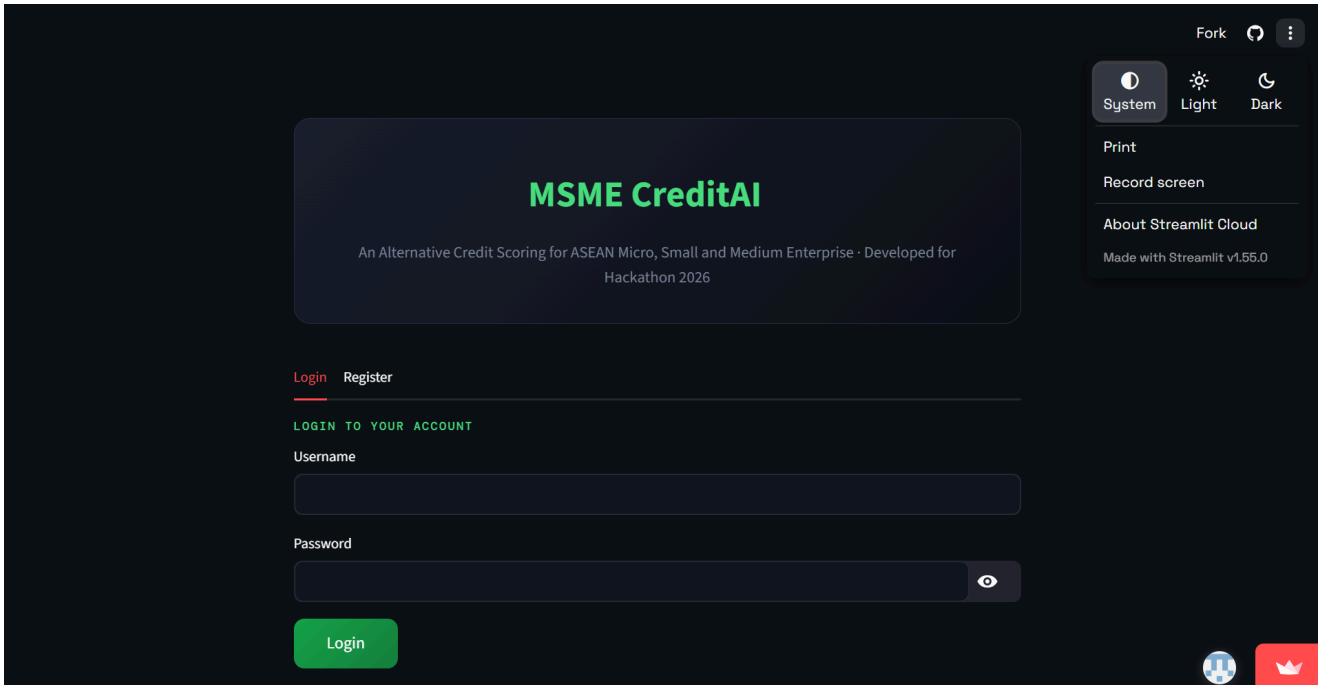



- When click the chosen “Deleted Selected Record”,

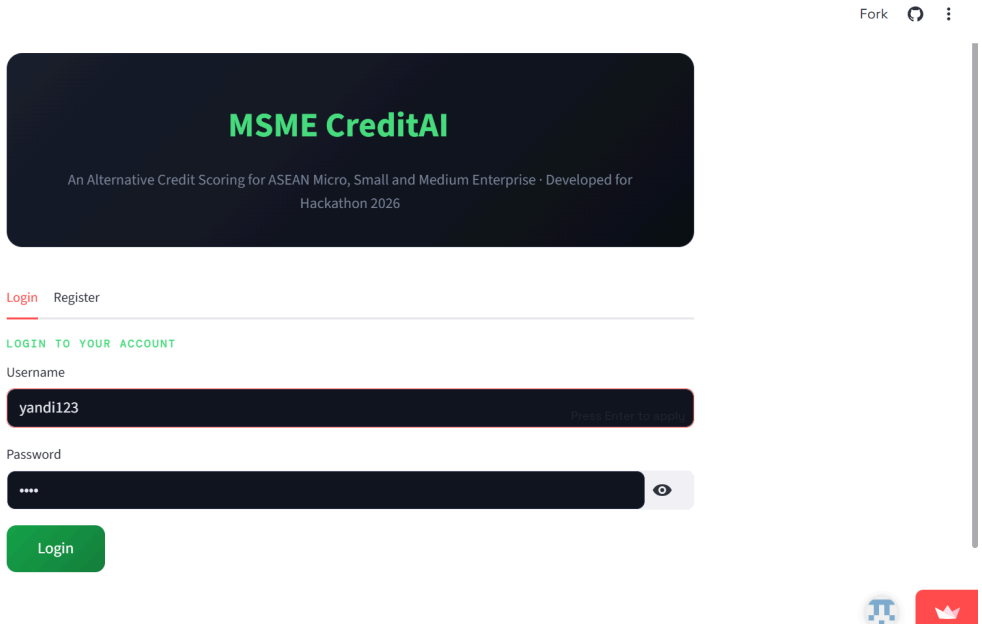
It will auto delete the record.



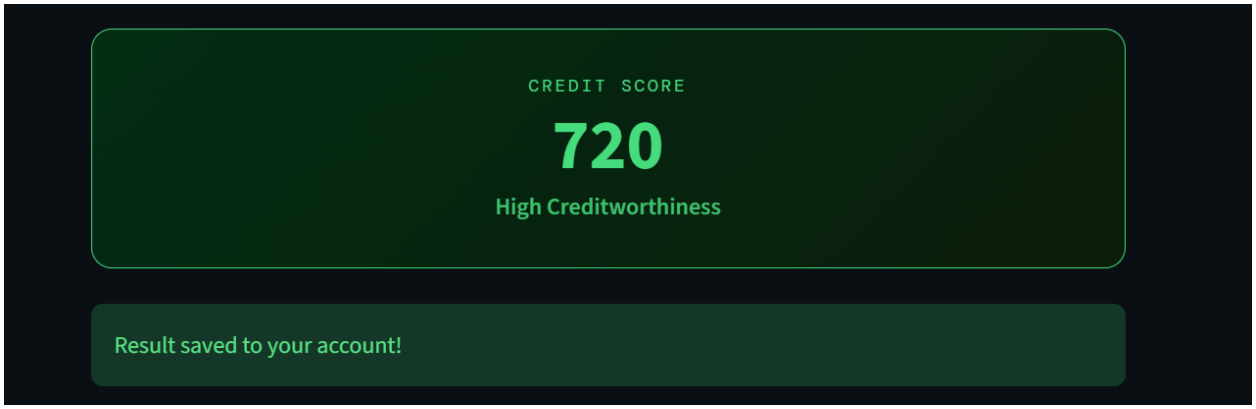
Users can customize their website theme by manually choosing the light mode, dark mode or following the system.



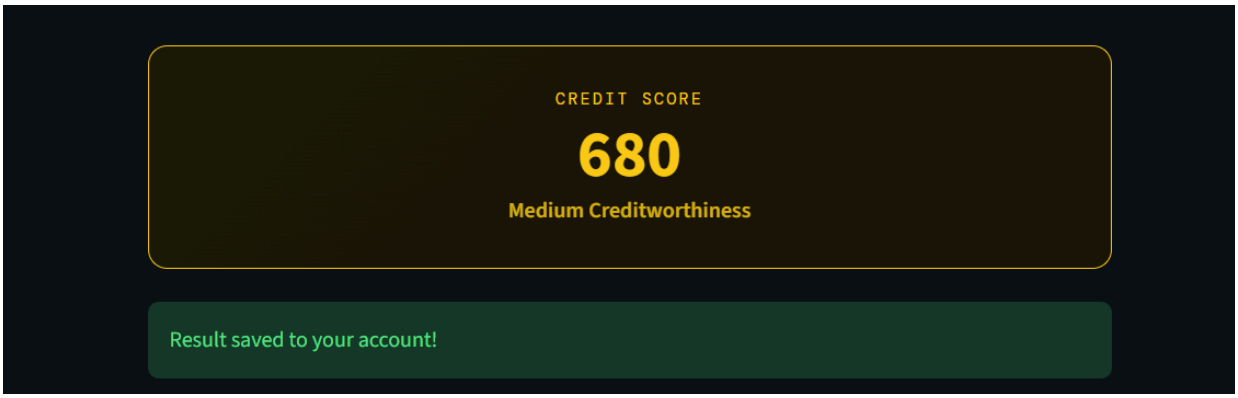

Here is the output that shows light mode.



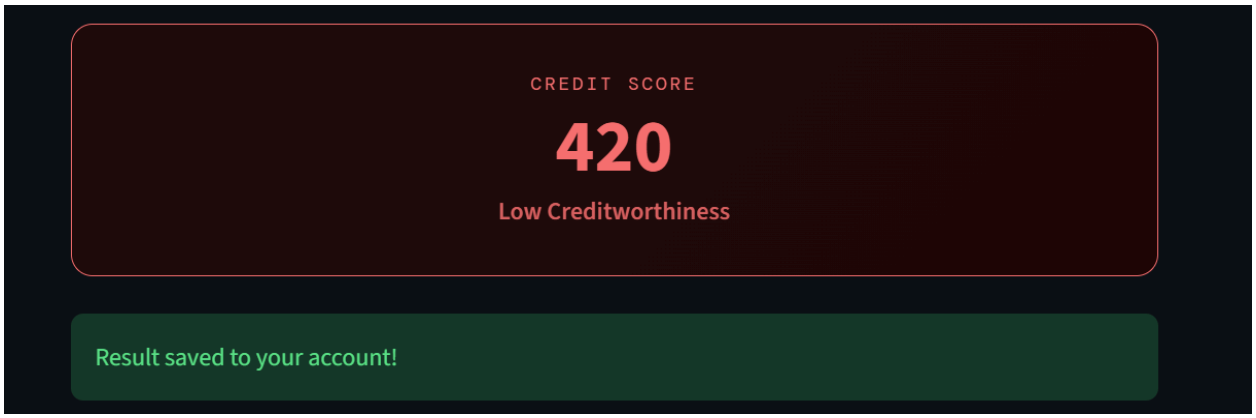
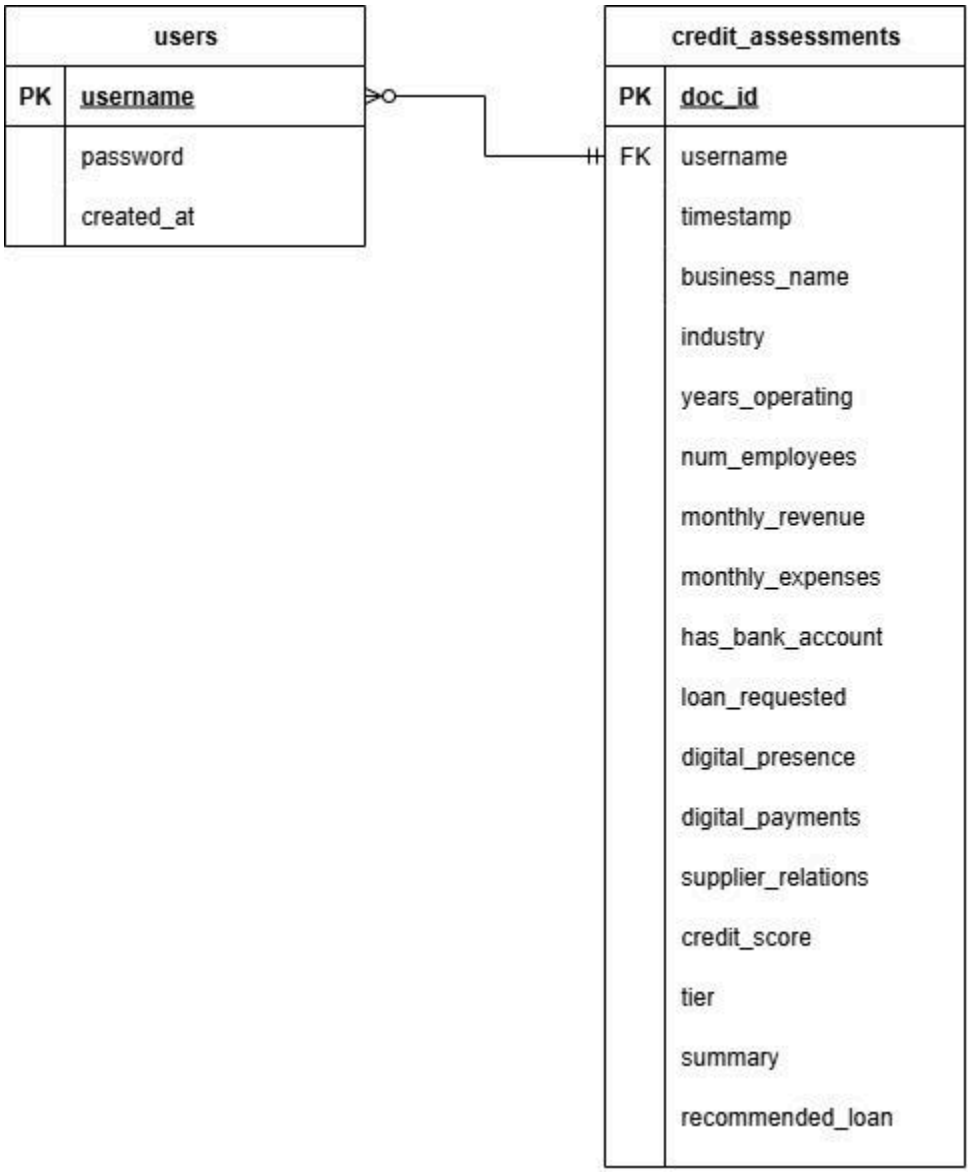
Green indicates that the company has a high creditworthiness.




Yellow indicates that the company has a moderate creditworthiness.




Red indicates that the company has a low creditworthiness.



## 9.0 Our Database Scheme